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1. If application is down in AWS – how to troubleshoot?

Follow a **step-by-step approach**:

Step 1: Check AWS Service Health

- Go to AWS Service Health Dashboard
- Verify if AWS itself has any outage

Step 2: Check EC2 Instance Status

- Is instance **running or stopped**?
- Check **Status Checks (2/2 passed)**

Step 3: Check Server Connectivity

- SSH into server:

`ssh ec2-user@your-ip`

- If not connecting → check:
 - Security Group (port 22 open)
 - Network ACL

Step 4: Check Application Process

`ps -ef | grep app`

- If not running → restart:

`sudo systemctl restart your-app`

Step 5: Check Logs

- Application logs:

`/var/log/`

- Web server logs:
 - Apache: `/var/log/httpd/`
 - Nginx: `/var/log/nginx/`

Step 6: Check Load Balancer (if used)

- Go to Elastic Load Balancing
- Verify:
 - Target group health
 - Instances are healthy

Step 7: Check Database

- If using Amazon RDS:
 - DB running?
 - Connection working?

2. Server reachable but application not working – troubleshooting

This is a **very common interview scenario**

Step 1: Verify Port

`netstat -tulnp | grep 80`

- Check if app is listening on correct port

Step 2: Restart Application

`sudo systemctl restart app-name`

Step 3: Check Application Logs

- Errors like:

- DB connection failed
- Memory issues
- Dependency errors

Step 4: Check Disk Space

df -h

- If full → app may stop

Step 5: Check CPU/Memory

top

- High usage → app may hang

Step 6: Check Security Groups

- Ensure required port (80/443/custom) is open

Step 7: Check Reverse Proxy

- If using Nginx/Apache:

sudo systemctl status nginx

Step 8: Check Environment Variables

- Missing configs → app won't start

3. Difference between CNAME and ALIAS

Feature	CNAME	ALIAS
Full form	Canonical Name	AWS-specific
Used for	Domain → Domain mapping	Domain → AWS resource
Root domain support	Not allowed	Allowed
Works with AWS services	No	Yes
Example	www.example.com → example.com	example.com → ALB/CloudFront

Key point:

- CNAME is **standard DNS record**
- ALIAS is **AWS Route 53 feature**

4. How Amazon Route 53 improves availability?

Route 53 improves availability using multiple routing strategies:

1. Health Checks

- Monitors endpoints (server/app)
- If unhealthy → traffic redirected automatically

2. Failover Routing

- Primary + Secondary setup
- If primary fails → switch to backup

3. Latency-Based Routing

- Routes users to **nearest server**
- Improves performance

4. Geolocation Routing

- Sends traffic based on user location

5. Multi-region Architecture

- Deploy app in multiple regions

- Route 53 distributes traffic

6. Integration with AWS Services

- Works with:
 - Amazon EC2
 - Elastic Load Balancing
 - Amazon CloudFront